

Interactive Sales Dashboard — Week 6

Data Visualization Mastery with Seaborn

Project Overview

This project analyzes sales data using Pandas, Seaborn, Matplotlib and Plotly. It covers sales trends, product performance, customer segmentation, statistical distributions and correlation analysis.

Objectives

- Apply Seaborn statistical visualization techniques.
- Create at least five chart types.
- Add interactive Plotly visualizations with hover information.
- Analyze customer segments and product performance.
- Combine charts into a professional dashboard layout.

Setup

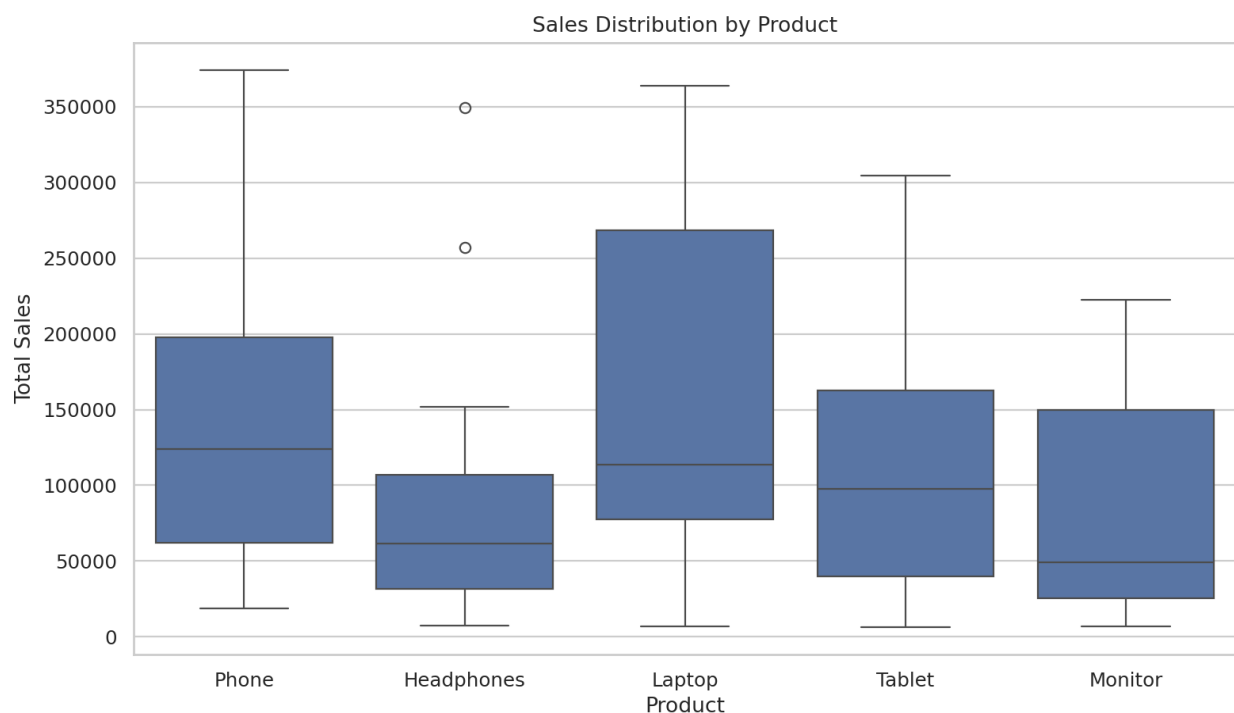
Install dependencies with `pip install -r requirements.txt`, then run `python dashboard.py`. Generated visuals are stored in the visualizations folder.

Code Structure

`dashboard.py` loads the CSV, preprocesses dates, creates statistical charts, summarizes products, calculates customer segments using sales percentiles, generates Plotly HTML charts, and creates the 2×2 dashboard overview.

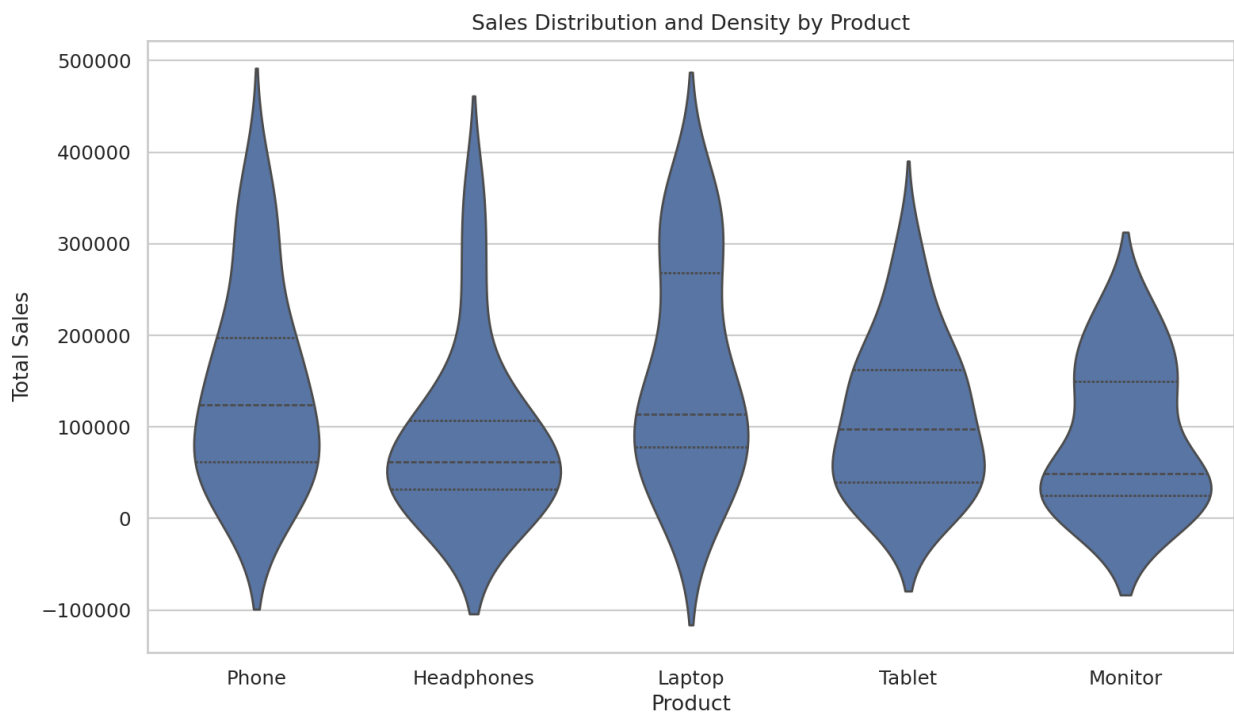
Box Plot

Compares sales distribution and variation across products.



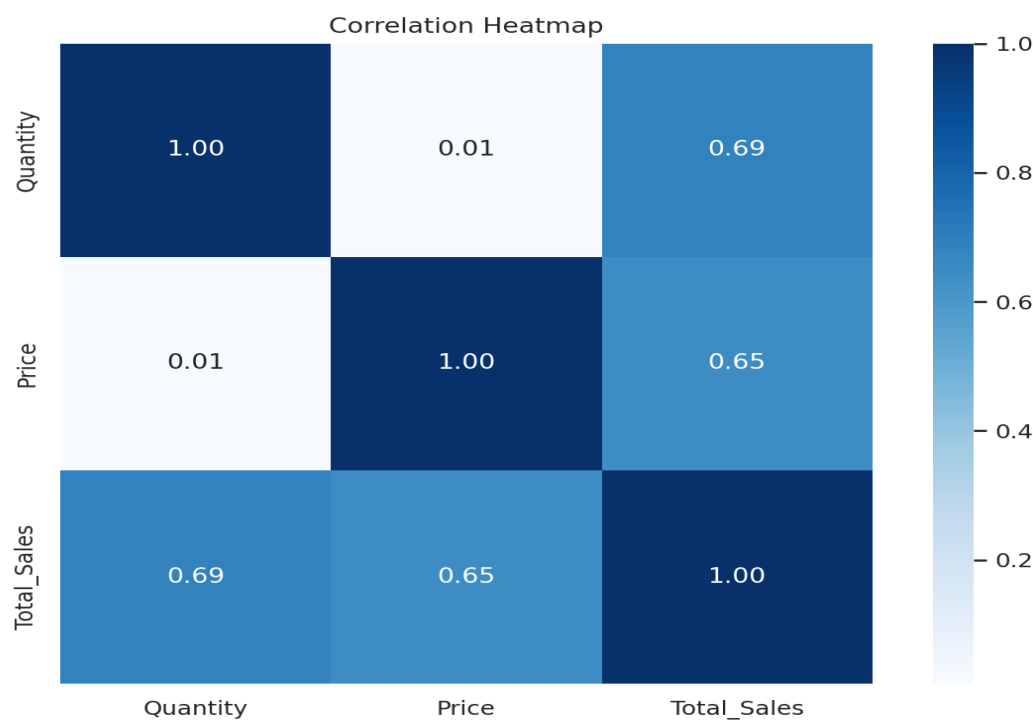
Violin Plot

Shows distribution density and quartile information across products.



Correlation Heatmap

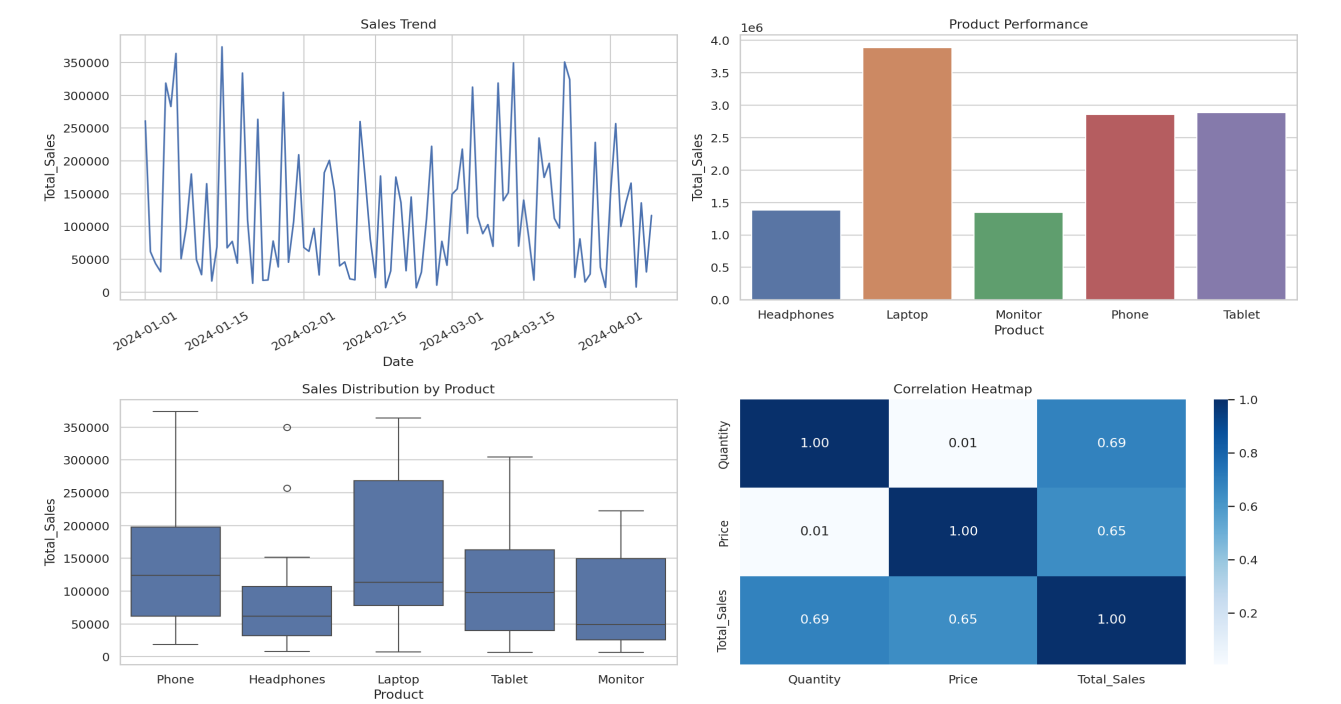
Shows relationships among numerical variables.



Dashboard Overview

Combines sales trend, product performance, box plot and correlation analysis in one 2x2 layout.

Interactive Sales Dashboard - Overview



Interactive Visualizations

The project includes interactive Plotly charts for sales trends, product performance, sales distribution and customer segmentation. Hover information, legends, zooming and panning allow users to explore the data.

Customer Segmentation Interpretation

Customers are divided into Low Value, Regular and High Value groups using the 33rd and 66th percentiles of customer total sales. The interactive scatter plot compares customer quantity with total sales and identifies each segment.

Technical Requirements Covered

- Seaborn statistical plots
- Seven chart types implemented
- Plotly interactive elements
- Cohesive visual styling
- Professional 2×2 dashboard layout
- Sales trends, customer segmentation and product performance
- Testing evidence

Testing

The dashboard script was executed successfully with a 100-row, 7-column sales dataset. All required static and interactive visualization files were generated successfully.